Contents

1 Symbols and Glossary

1.1 Key Terms and Definitions

1.1.1 Olfaction and Chemosensation

- **Olfaction**: The sense of smell; the ability to detect and identify airborne molecules through specialized sensory organs
- **Chemosensation**: The detection of chemical stimuli by sensory cells, including olfaction, gustation, and chemesthesis

- **Semiochemicals**: Chemical substances that carry information between organisms, including pheromones, allomones, and kairomones
- **Pheromones**: Semiochemicals that affect the behavior of other members of the same species, such as sex pheromones and trail pheromones
- **Cuticular Hydrocarbons (CHCs)**: Long-chain hydrocarbons found on the surface of insects that serve as recognition cues and waterproofing agents
- **Sensilla**: Microscopic sensory hairs or pegs on insect antennae and other body parts that serve as the primary sensory units for olfaction and other senses

1.1.2 Insect Anatomy and Physiology

- **Antennae**: Paired sensory appendages on the head of insects that contain olfactory and other sensory receptors
- **Sensilla**: Microscopic sensory hairs or pegs on insect antennae and other body parts that serve as the primary sensory units
- **Sensilla Trichodea**: Hair-like sensilla that are often involved in olfaction, typically 6-160 μm in length
- **Sensilla Basiconica**: Peg-like sensilla with porous surfaces, typically 2-8 μm in length
- **Sensilla Coeloconica**: Pit-like sensilla that may detect temperature, humidity, and infrared radiation
- **ORN**: Olfactory Receptor Neuron; nerve cells that respond to chemical stimuli and transmit signals to the brain
- **OR**: Olfactory Receptor; membrane proteins that bind to odor molecules and initiate signal transduction
- **Antennal Lobe (AL)**: First olfactory processing center in the insect brain containing glomeruli that aggregate ORN inputs by receptor type
- **Glomerulus (plural: glomeruli)**: Spheroidal neuropil compartment in the AL where ORN axons synapse with projection neurons and local interneurons; often tuned to receptor families or vibrational features

1.1.3 Electromagnetic Theory and Infrared Detection

- **Infrared (IR)**: Electromagnetic radiation with wavelengths longer than visible light (0.7-1000 μm), invisible to human eyes but detectable by specialized sensors
- **Mid-infrared (MIR)**: IR radiation in the 2-25 μm range, corresponding to molecular vibrational modes and fundamental for chemical sensing applications
- **Far-infrared (FIR)**: IR radiation in the 25-1000 μm range, corresponding to rotational and low-frequency vibrational modes, also known as thermal infrared
- **Near-infrared (NIR)**: IR radiation in the 0.7-2 μm range, just beyond visible light, commonly used in spectroscopy and optical communications
- **Dielectric**: A material that can be polarized by an electric field and supports electromagnetic wave propagation
- **Waveguide**: A structure that guides electromagnetic waves along a specific path with minimal loss
- **Resonator**: A device or structure that oscillates at specific frequencies, amplifying signals at resonant frequencies
- **Quality Factor (Q)**: A measure of resonator performance, defined as the ratio of stored energy to

energy lost per cycle

1.1.4 Spectroscopy and Molecular Properties

- **Vibrational Theory**: The hypothesis that olfaction works by detecting molecular vibrations in the infrared spectrum rather than molecular shape, providing a mechanism for odor recognition at the quantum mechanical level
- **Emission Spectrum**: The range of wavelengths of electromagnetic radiation emitted by a substance when excited, characteristic of the energy level transitions in the material
- **Absorption Spectrum**: The range of wavelengths absorbed by a substance, complementary to emission spectra and determined by the molecular structure and bonding
- **Transmission Window**: A range of wavelengths where the atmosphere is relatively transparent to electromagnetic radiation, allowing for long-range signal propagation
- **Deuteration**: The replacement of hydrogen atoms with deuterium (heavy hydrogen) in molecules, affecting vibrational frequencies
- **Enantiomers**: Mirror-image forms of the same molecule that may have different olfactory properties
- **FRET**: Förster Resonance Energy Transfer; energy transfer between molecules through dipole-dipole interactions
- **Wavenumber**: The reciprocal of wavelength, typically expressed in cm^{-1}, related to energy by $E = hc\tilde{\nu}$

1.2 Mathematical Notation

1.2.1 Wavelength and frequency

- λ **(lambda)**: Wavelength, typically in micrometers (μm) or nanometers (nm).
- ν **(nu)**: Frequency in Hz, related to wavelength by $c = \lambda\nu$.
- $\tilde{\nu}$ **(wavenumber)**: Reciprocal wavelength in cm^{-1}, $\tilde{\nu} = 10^4/\lambda$ (for λ in μm).
- **c**: Speed of light in vacuum (2.998 × 10^8 m/s).
- **μm**: Micrometer (10^-6 m); standard unit for infrared wavelengths.
- **nm**: Nanometer (10^-9 m).
- **cm^-1**: Wavenumber unit used in IR spectroscopy.

1.2.2 Physical Constants and Units

- **h**: Planck's constant (6.626 × 10^-34 J · s)
- $\hbar$: Reduced Planck constant (h/2π = 1.055 × 10^-34 J · s)
- **k_B**: Boltzmann constant (1.381 × 10^-23 J/K)
- **T**: Temperature in Kelvin (K)
- ϵ**_0**: Permittivity of free space (8.854 × 10^-12 F/m)
- μ**_0**: Permeability of free space (4π × 10^-7 H/m)
- **e**: Elementary charge (1.602 × 10^-19 C)

1.2.3 Electromagnetic Theory

- **E**: Electric field vector (V/m)
- **B**: Magnetic induction vector (T)
- **D**: Electric displacement field ($\mathrm{C/m^2}$)
- **H**: Magnetic field vector (A/m)
- **P**: Polarization vector ($\mathrm{C/m^2}$)

- **M**: Magnetization vector (A/m)
- ϵ__**r**: Relative permittivity (dimensionless)
- μ__**r**: Relative permeability (dimensionless)
- **tan** δ: Loss tangent, measure of dielectric loss (dimensionless)

1.2.4 Insect Measurements and Response Times

- μ**m**: Micrometer; typical size range for insect sensilla (1-200 μm)
- **nm**: Nanometer; scale of molecular interactions and receptor dimensions
- **ms**: Millisecond; typical response time of insect ORNs (1-5 ms)
- μ**s**: Microsecond; time scale for electromagnetic detection
- **ns**: Nanosecond; time scale for quantum processes

1.3 Abbreviations and Acronyms

1.3.1 General Scientific Terms

- **OR**: Olfactory Receptor
- **ORNs**: Olfactory Receptor Neurons
- **CHCs**: Cuticular Hydrocarbons
- **GPCR**: G-Protein Coupled Receptor
- **MTs**: Microtubules
- **FRET**: Förster Resonance Energy Transfer
- **SNR**: Signal-to-Noise Ratio
- **Q**: Quality Factor
- **ROC**: Receiver Operating Characteristic

1.3.2 Infrared and Spectroscopy

- **IR**: Infrared
- **FIR**: Far Infrared
- **MIR**: Mid Infrared
- **NIR**: Near Infrared
- **ATR-FTIR**: Attenuated Total Reflectance Fourier Transform Infrared Spectroscopy
- **FTIR**: Fourier Transform Infrared Spectroscopy
- **Raman**: Raman Spectroscopy
- **UV-Vis**: Ultraviolet-Visible Spectroscopy

1.3.3 Computational and Analytical

- **API**: Application Programming Interface
- **TDD**: Test-Driven Development
- **MAE**: Mean Absolute Error
- **RMSE**: Root Mean Square Error
- **ANOVA**: Analysis of Variance
- **PER**: Proboscis Extension Reflex

1.4 Key Concepts and Relationships

1.4.1 Atmospheric Transmission Windows

The Earth's atmosphere has specific wavelength ranges where infrared radiation can travel relatively freely, enabling long-range detection of insect semiochemicals:

- **2-5 μm (Mid-infrared)**: ~80% transmission efficiency, optimal for hydrocarbon detection
- **8-14 μm (Long-wave infrared)**: ~90% transmission efficiency, optimal for long-range communication; ground materials can emit infrared energy that partially penetrates this window
- **17-25 μm (Far-infrared)**: ~70% transmission efficiency, useful for thermal detection

Transmission function: Modeled by `src/core.py::calculate_atmospheric_transmission()` with validation against peer-reviewed atmospheric physics studies (see (**??**); unit tests in `tests/test_core.py`):

$$T(\lambda) = \exp\left[-\sum_i \alpha_i(\lambda) L_i\right] \tag{1}$$

where $\alpha_i(\lambda)$ is the absorption coefficient and L_i is the path length through atmospheric component i.

1.4.2 Sensilla Dimensions and Wavelength Matching

Insect sensilla have dimensions that correspond closely to the wavelengths of infrared radiation they detect, as confirmed by comparative morphometric studies:

- **Sensilla Trichodea**: 6-160 μm length, optimal for 2-30 μm wavelengths
- **Sensilla Basiconica**: 2-8 μm length, optimal for 1-10 μm wavelengths; specific dimensions of 6.86–53.42 μm observed in thrips species
- **Sensilla Coeloconica**: 5-15 μm length, optimal for 3-20 μm wavelengths
- **Specialized IR organs**: Approximately 100 sensilla per organ in beetle species

Wavelength matching: Analyzed by `src/sensilla.py::analyze_sensilla_dimensions()` with validation against peer-reviewed morphometric data; see resonant frequency (2) and tests `tests/test_sensilla.py`. Publication figures are generated via `scripts/generate_research_figures.py`.

Resonant Frequency: The fundamental resonant frequency of a sensillum is:

$$f_{res} = \frac{c}{2\pi}\sqrt{\left(\frac{\alpha_{mn}}{a}\right)^2 + \left(\frac{p\pi}{L}\right)^2} \tag{2}$$

where c is the speed of light, α_{mn} is the Bessel function root, and a and L are the radius and length.

1.4.3 Response Time Comparisons

Different sensory modalities exhibit characteristic response times that reflect their underlying mechanisms:

- **Insect ORNs (Infrared)**: 1-5 ms response time
- **Insect Photoreceptors**: 0.1 ms response time
- **Insect Auditory Receptors**: 0.16 ms response time

- **Traditional Olfaction (Molecular)**: 7-12 ms response time
- **Mammalian ORNs**: 10-50 ms response time

Response time analysis: Compared using `src/core.py::calculate_response_time_improvement()`; see `tests/test_core.py::TestResponseTimeImprovement`. See `figures/response_time_comparison.png` and cf. (**??**).

1.4.4 Signal Processing and Information Theory

The vibrational theory incorporates advanced signal processing concepts:

- **Channel Capacity**: The maximum information rate that can be transmitted through the infrared detection channel:

$$C = B \log_2(1 + SNR) \tag{3}$$

where B is the bandwidth and SNR is the signal-to-noise ratio.

- **Detection Threshold**: The minimum detectable power is:

$$P_{min} = k_B T \Delta f \cdot SNR_{min} \tag{4}$$

where k_B is Boltzmann's constant, T is temperature, Δf is bandwidth, and SNR_{min} is the minimum required signal-to-noise ratio.

1.5 Research Methodology Terms

1.5.1 Experimental Techniques

- **Ionotropic**: Direct ligand-gated ion channels that open immediately upon binding
- **Metabotropic**: G-protein coupled receptor systems that activate intracellular signaling cascades
- **Behavioral Conditioning**: Training insects to associate specific stimuli with rewards or punishments
- **Electroantennography (EAG)**: Recording electrical responses from insect antennae to chemical stimuli
- **Single Sensillum Recording**: Recording from individual sensilla to measure response characteristics

1.5.2 Physical and Chemical Properties

- **Piezoelectric**: Materials that generate electric charge in response to mechanical stress
- **Allosteric Modulation**: Changes in protein function due to binding at sites other than the active site
- **Photomodulation**: Changes in protein function due to light absorption
- **Dielectric Loss**: Energy dissipation in dielectric materials due to molecular motion
- **Resonant Coupling**: Enhanced energy transfer when systems oscillate at the same frequency

1.5.3 Statistical and Analytical Methods

- **Power Analysis**: Statistical method to determine the minimum sample size needed to detect an effect
- **Receiver Operating Characteristic (ROC)**: Plot of true positive rate vs. false positive rate
- **Discriminability Index (d')**: Measure of ability to distinguish between signal and noise
- **Hill Coefficient**: Measure of cooperativity in binding or response functions
- **Log-Periodic Analysis**: Analysis of systems with periodic spacing that increases logarithmically

1.6 Source Code Implementation

All mathematical concepts and equations presented in this manuscript are implemented in tested source code that generates the visualizations and analyses embedded throughout. The key functions include:

1.6.1 Core Physics and Calculations

- `calculate_atmospheric_transmission()`: Implements atmospheric transmission models with environmental parameter integration
- `calculate_response_time_improvement()`: Compares response times across different sensory modalities with statistical validation
- `calculate_wavelength_from_wavenumber()`: Converts between wavelength and wavenumber representations
- `safe_division()`: Performs safe division operations with error handling

1.6.2 Morphological and Structural Analysis

- `analyze_sensilla_dimensions()`: Analyzes sensilla morphology and calculates optimal detection wavelengths
- `calculate_sensilla_resonance_frequency()`: Computes resonant frequencies using cavity resonator theory
- `calculate_wavelength_matching()`: Quantifies wavelength matching between sensilla and incident radiation
- `generate_sensilla_visualization()`: Creates detailed visualizations of sensilla structures and properties

1.6.3 Spectroscopic and Chemical Analysis

- `analyze_chc_spectra()`: Processes cuticular hydrocarbon spectroscopic data with peak detection
- `calculate_spectral_overlap()`: Quantifies spectral similarity between different compounds
- `generate_spectral_plots()`: Creates publication-quality spectral visualizations
- `identify_chc_compounds()`: Identifies potential CHC compounds based on peak positions

1.6.4 Behavioral and Response Analysis

- `analyze_behavioral_response()`: Analyzes behavioral response data with statistical testing
- `calculate_power_analysis()`: Performs statistical power analysis for experimental design
- `calculate_response_statistics()`: Computes comprehensive statistics for response data
- `generate_behavioral_plots()`: Creates behavioral response visualizations

1.6.5 Integrated Analysis Frameworks

- `IntegratedAnalyzer`: Combines multiple analytical approaches for comprehensive assessment

- **`MetaMaterialAnalyzer`**: Analyzes meta-material properties and quantum effects
- **`FermiEstimator`**: Performs Fermi estimation for order-of-magnitude calculations
- **`BehavioralAnalyzer`**: Specialized analysis for behavioral response data

1.6.6 Data Validation and Testing

- **`validate_numeric_inputs()`**: Ensures all numeric inputs are valid and finite (exercised in multiple unit tests)
- **`SensillaData`**: Container class for sensilla measurements with validation
- **`SpectralData`**: Container class for spectral data with analysis methods
- **`BehavioralData`**: Container class for behavioral data with statistical analysis

1.7 References and Further Reading

For detailed discussions of the concepts presented here, see:

- **Electromagnetic Theory**: Insect sensilla as dielectric waveguides
- **Spectroscopic Analysis**: Cuticular hydrocarbon spectroscopy
- **Quantum Models**: Electron tunneling in olfactory reception
- **Behavioral Analysis**: Response time comparisons
- **Atmospheric Transmission**: Infrared transmission characteristics

1.8 Computational Framework Documentation

The complete computational framework is documented with (appendix case studies: sec:app$_{s}ensilla_{a}rray, sec : app$

- **100% Test Coverage**: All functions are tested with comprehensive unit and integration tests
- **Performance Benchmarks**: Execution speed and memory efficiency metrics
- **Validation Procedures**: Comparison with known physical constants and empirical data
- **API Documentation**: Complete function signatures and parameter descriptions
- **Example Scripts**: Demonstrations of complete analysis pipelines

For complete mathematical formulations and source code implementation, see Section sec:mathematical$_{a}ppendix.Cr$
$linkstoimplementationsandunittestsareincludedtherein$.

Module	Name	Kind	Summary
`__init__`	`get_package_info`	function	Get comprehensive package information
`__init__`	`run_demo_analysis`	function	Run a demonstration analysis using all available frameworks
`ant_stack.antbody`	`AntBodySensilla`	class	Sensilla configuration using cohereAnts morphology analysis
`ant_stack.antbody`	`AntBodySpectroscopy`	class	Atmospheric transmission access aligned with core calculations

Module	Name	Kind	Summary
`ant_stack.antbrain`	`AntBrainOlfaction`	class	High-level olfactory pipeline stub with AL→MB→CX placeholders
`ant_stack.antbrain`	`VibrationalGlomeruliCircuit`	class	Bank of resonant channels tuned across 2–25 μm
`ant_stack.antmind`	`AntMindOlfaction`	class	Active-inference-like placeholder for olfactory policy selection
`ant_stack.antmind`	`AntMindStigmergy`	class	Grid-based pheromone field with diffusion and decay
`behavioral`	`BehavioralAnalyzer`	class	Main analyzer for behavioral response data
`behavioral`	`BehavioralData`	class	Container for behavioral response data with validation
`behavioral`	`StatisticalAnalyzer`	class	Statistical analysis for behavioral data
`behavioral`	`analyze_behavioral_response`	function	Analyze behavioral response data
`behavioral`	`calculate_power_analysis`	function	Calculate statistical power for the comparison
`behavioral`	`calculate_response_statistics`	function	Calculate comprehensive statistics for behavioral response data
`behavioral`	`generate_behavioral_plots`	function	Generate behavioral response plots
`case_studies.active_inference`	`olfactory_active_inference_step`	function	Minimal deterministic update step for a 2D position under a gradient cue
`case_studies.detection_limits`	`detection_performance_vs_snr`	function	Analyze detection performance vs signal-to-noise ratio
`case_studies.detection_limits`	`detection_range_analysis`	function	Analyze detection range for IR olfactory communication

Module	Name	Kind	Summary
`case_studies.detection_limits`	`min_detectable_power`	function	Minimum detectable signal power using thermal noise floor and SNR threshold
`case_studies.detection_limits`	`noise_floor_analysis`	function	Analyze noise floor components vs frequency
`case_studies.detection_limits`	`operating_point`	function	Bundle operating point parameters deterministically
`case_studies.detection_limits`	`operating_regions_analysis`	function	Analyze operating regions in power-temperature space
`case_studies.detection_limits`	`optimize_detection_parameters`	function	Optimize detection system parameters for given constraints and objectives
`case_studies.detection_limits`	`roc_analysis`	function	Receiver Operating Characteristic (ROC) analysis for signal detection
`case_studies.detection_limits`	`sensitivity_analysis`	function	Sensitivity analysis of detection performance to parameter variations
`case_studies.detection_limits`	`snr_curve`	function	SNR vs
`case_studies.environmental_channel`	`atmospheric_transmission_comprehensive`	function	Comprehensive atmospheric transmission model with multiple physical effects
`case_studies.environmental_channel`	`atmospheric_transmission_detailed`	function	Compute a simple parametric atmospheric transmission curve
`case_studies.environmental_channel`	`channel_capacity_analysis`	function	Analyze communication channel capacity under atmospheric conditions
`case_studies.environmental_channel`	`channel_capacity_vs_env`	function	Map Shannon capacity across humidity×temperature grid (legacy function)

Module	Name	Kind	Summary
`case_studies.environmental_channel`	`environmental_sensitivity_analysis`	function	Analyze sensitivity of transmission to environmental parameters
`case_studies.environmental_channel`	`molecular_absorption_cross_section`	function	Calculate molecular absorption cross-sections for atmospheric constituents
`case_studies.environmental_channel`	`optimize_wavelength_for_range`	function	Find optimal wavelengths for target communication range and capacity
`case_studies.environmental_channel`	`rayleigh_scattering_coefficient`	function	Calculate Rayleigh scattering coefficient for dry air
`case_studies.neural_encoding`	`adaptation_dynamics_analysis`	function	Analyze adaptation dynamics in neural responses
`case_studies.neural_encoding`	`analyze_spike_train_statistics`	function	Compute comprehensive spike train statistics
`case_studies.neural_encoding`	`generate_spike_trains`	function	Generate realistic spike trains for ORN responses to odor stimuli
`case_studies.neural_encoding`	`information_rate_time_series`	function	Estimate information metrics using a Gaussian channel approximation
`case_studies.neural_encoding`	`mutual_information_analysis`	function	Compute mutual information between neural responses and stimuli
`case_studies.neural_encoding`	`odor_discrimination_analysis`	function	Analyze odor discrimination performance across different time windows
`case_studies.neural_encoding`	`population_coding_analysis`	function	Analyze population coding efficiency across multiple ORNs

Module	Name	Kind	Summary
case_studies.neural_encoding	rate_coding_metrics	function	Compute simple separability metrics (means/stds) deterministically
case_studies.neural_encoding	temporal_coding_analysis	function	Analyze temporal coding precision and response latency
case_studies.plasmonic_geometry	coupled_dipoles_near_field	function	Calculate near-field enhancement for coupled plasmonic nanoparticles
case_studies.plasmonic_geometry	drude_model_permittivity	function	Calculate frequency-dependent permittivity using Drude model
case_studies.plasmonic_geometry	field_distribution_near_particle	function	Calculate near-field distribution around a spherical nanoparticle
case_studies.plasmonic_geometry	mie_scattering_sphere	function	Calculate Mie scattering properties for spherical nanoparticles
case_studies.plasmonic_geometry	optimize_plasmonic_geometry	function	Optimize nanoparticle geometry for maximum enhancement at target wavelength
case_studies.plasmonic_geometry	sweep_plasmonic_quality	function	Comprehensive sweep of plasmonic quality factors across size and wavelength
case_studies.sensilla_array_directionality	analyze_sensilla_morphology	function	Analyze sensilla dimensions for resonant wavelength matching
case_studies.sensilla_array_directionality	array_gain	function	Compute a scalar array gain proxy as peak-to-mean power ratio
case_studies.sensilla_array_directionality	array_pattern_2d	function	Compute 2D radiation pattern for sensilla array across frequency range
case_studies.sensilla_array_directionality	compute_beam_pattern	function	Compute a simplified 1D beam pattern over wavelengths

Module	Name	Kind	Summary
case_studies.sensilla_array_directionality	design_circular_array	function	Design a circular antenna array representing sensilla on insect antennae
case_studies.sensilla_array_directionality	design_log_periodic_array	function	Design a 1D log-periodic array of element positions
case_studies.sensilla_array_directionality	frequency_response_analysis	function	Analyze frequency response characteristics of sensilla array
case_studies.sensilla_array_directionality	mutual_coupling_matrix	function	Compute mutual coupling matrix between antenna elements
case_studies.sensilla_array_directionality	single_element_pattern	function	Individual sensillum radiation pattern as function of observation angle
case_studies.spectral_unmixing	advanced_classification_suite	function	Comprehensive classification analysis using multiple algorithms
case_studies.spectral_unmixing	generate_realistic_chc_spectra	function	Generate realistic CHC spectral data with known ground truth components
case_studies.spectral_unmixing	independent_component_analysis_spectra	function	Independent Component Analysis (ICA) for blind source separation of spectra
case_studies.spectral_unmixing	lda_baseline	function	Closed-form two-class LDA with equal covariance; returns accuracy on train
case_studies.spectral_unmixing	nmf_unmix	function	Deterministic, simple NMF via multiplicative updates
case_studies.spectral_unmixing	performance_metrics_comprehensive	function	Compute comprehensive performance metrics for classification
case_studies.spectral_unmixing	spectral_feature_extraction	function	Extract discriminative features from spectral data

Module	Name	Kind	Summary
case_studies.spectral_unmixing	vertex_component_analysis	function	Vertex Component Analysis (VCA) for endmember extraction
config	ConfigManager	class	Centralized configuration manager for insect analysis
config	enable_verbose_logging	function	Enable verbose logging for debugging
config	get_config	function	Get the global configuration manager instance
config	init_config	function	Initialize the global configuration manager
config	set_plot_style	function	Set matplotlib plot style
config	set_random_seed	function	Set random seed for reproducible results
config	set_temperature	function	Set analysis temperature in Kelvin
core	calculate_atmospheric_transmission	function	Calculate atmospheric transmission for given wavelengths in the infrared spectrum
core	calculate_response_time_improvement	function	Calculate the improvement in response time compared to traditional olfaction
core	calculate_wavelength_from_wavenumber	function	Convert wavenumber (cm^{1}) to wavelength (μm)
core	calculate_wavenumber_from_wavelength	function	Convert wavelength (μm) to wavenumber (cmˆ-1)
core	safe_division	function	Safely perform division, returning infinity if denominator is zero
core	validate_numeric_inputs	function	Validate that all numeric inputs are finite numbers
fermi_estimation	FermiEstimator	class	Comprehensive Fermi Estimation analyzer for olfaction and infrared sensing

Module	Name	Kind	Summary
`fermi_estimation`	`create_sample_fermi_analysis`	function	Create a sample Fermi analysis for demonstration
`glossary_gen`	`ApiEntry`	class	Represents a public API entry from source code
`glossary_gen`	`build_api_index`	function	Scan `src_dir` and collect public functions/classes with summaries
`glossary_gen`	`generate_markdown_table`	function	Generate a Markdown table from API entries
`glossary_gen`	`inject_between_markers`	function	Replace content between begin_marker and end_marker (inclusive markers preserved)
`insect_analysis`	`run_comprehensive_analysis`	function	Run comprehensive analysis using all available frameworks
`integrated_analysis`	`IntegratedAnalyzer`	class	Integrated analyzer combining Fermi Estimation and meta-material frameworks
`integrated_analysis`	`create_sample_integrated_analysis`	function	Create a sample integrated analysis for demonstration
`meta_material_framework`	`MetaMaterialAnalyzer`	class	Comprehensive meta-material analyzer for olfaction and infrared sensing
`meta_material_framework`	`create_sample_metamaterial_analysis`	function	Create a sample meta-material analysis for demonstration
`sensilla`	`SensillaData`	class	Container for sensilla measurement data with validation
`sensilla`	`analyze_sensilla_dimensions`	function	Analyze sensilla dimensions and calculate optimal detection wavelengths

Module	Name	Kind	Summary
`sensilla`	`calculate_sensilla_resonance_frequency`	function	Calculate the fundamental resonance frequency of a sensillum
`sensilla`	`calculate_wavelength_matching`	function	Calculate wavelength matching between sensilla dimensions and incident radiation
`sensilla`	`generate_sensilla_visualization`	function	Generate a visualization of sensilla dimensions and optimal wavelengths
`spectroscopy`	`CHCAnalyzer`	class	Analyzer for cuticular hydrocarbon spectra
`spectroscopy`	`PeakFinder`	class	Peak detection and analysis for spectral data
`spectroscopy`	`SpectralData`	class	Container for spectral data with validation and analysis methods
`spectroscopy`	`analyze_chc_spectra`	function	Analyze cuticular hydrocarbon (CHC) infrared spectra
`spectroscopy`	`calculate_spectral_overlap`	function	Calculate spectral overlap between two spectra
`spectroscopy`	`generate_spectral_plots`	function	Generate spectral plots for multiple compounds
`spectroscopy`	`identify_chc_compounds`	function	Identify potential CHC compounds based on peak positions
`visualization`	`AdvancedVisualizer`	class	Advanced visualization tools for insect analysis data
`visualization`	`PlotStyler`	class	Advanced plot styling and theming system
`visualization`	`create_publication_figure`	function	Create a publication-ready figure with optimal styling
`visualization`	`create_subplots`	function	Create subplots with consistent styling
`visualization`	`get_colorblind_palette`	function	Get a colorblind-friendly color palette

Module	Name	Kind	Summary
visualization	set_plot_style	function	Set the global plot style